

## EDUCATION

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### University of Toronto

Toronto, Canada

Doctor of Philosophy (PhD), Aerospace Science and Engineering

2018 – 2024

**Thesis:** Provably stable discontinuous spectral-element methods with the summation-by-parts property: Unified matrix analysis and efficient tensor-product formulations on curved simplices

**Advisor:** David Zingg

### Carleton University

Ottawa, Canada

Bachelor of Engineering (BEng), Mechanical Engineering

2014 – 2018

**Capstone project:** Performance analysis and numerical optimization of a supercritical carbon dioxide Brayton cycle

## RESEARCH INTERESTS

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Structure-preserving numerical methods for nonlinear partial differential equations; efficient algorithms for high-performance scientific computing; simulation software development; scientific and engineering applications including aerodynamics, plasma physics, and earth system modelling (i.e., atmospheric, hydrologic, and ocean simulations)

## PUBLICATIONS IN PEER-REVIEWED JOURNALS

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T. Montoya, A. M. Rueda-Ramírez, and G. J. Gassner, [Entropy-stable discontinuous spectral-element methods for the spherical shallow water equations in covariant form](#). *Journal of Computational Physics* 555, article no. 114782, 2026.

T. Montoya and D. W. Zingg, [Efficient entropy-stable discontinuous spectral-element methods using tensor-product summation-by-parts operators on triangles and tetrahedra](#). *Journal of Computational Physics* 516, article no. 113360, 2024.

T. Montoya and D. W. Zingg, [Efficient tensor-product spectral-element operators with the summation-by-parts property on curved triangles and tetrahedra](#). *SIAM Journal on Scientific Computing* 46(4), pp. A2270–A2297, 2024.

T. Montoya and D. W. Zingg, [A unifying algebraic framework for discontinuous Galerkin and flux reconstruction methods based on the summation-by-parts property](#). *Journal of Scientific Computing* 92(3), article no. 87, 2022.

## PUBLICATIONS IN CONFERENCE PROCEEDINGS

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T. Montoya and D. W. Zingg, [Stable and conservative high-order methods on triangular elements using tensor-product summation-by-parts operators](#). 11<sup>th</sup> *International Conference on Computational Fluid Dynamics*, Maui, United States, July 2022. **Awarded best student paper.**

## CONFERENCE PRESENTATIONS (EXCLUDING ABOVE PROCEEDINGS)

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**Actor-based orchestration for heterogeneous simulation workflows.** *Canadian Symposium on Fluid Dynamics*, Regina, Canada, June 2026.

**Discontinuous spectral-element methods for hyperbolic systems on manifolds.** *Canadian Applied and Industrial Mathematics Society Annual Meeting*, Regina, Canada, June 2026.

**Entropy-stable discontinuous Galerkin methods for the spherical shallow water equations in flux form.** *International Conference on Spectral and High-Order Methods*, Montréal, Canada, July 2025.

**Entropy-stable discontinuous spectral-element methods for the shallow water equations on the sphere.** *Southern Ontario Numerical Analysis Day*, Hamilton, Canada, May 2025.

**Efficient entropy-stable discontinuous spectral-element methods in collapsed coordinates for hyperbolic systems on curved triangular and tetrahedral meshes.** *Canadian Applied and Industrial Mathematics Society Annual Meeting*, Kingston, Canada, June 2024.

**Efficient entropy-stable tensor-product spectral-element methods on simplices.** 9<sup>th</sup> *European Congress on Computational Methods in Applied Sciences and Engineering*, Lisbon, Portugal, June 2024.

**Efficient tensor-product spectral-element methods with the summation-by-parts property on triangles and tetrahedra.** *SIAM Conference on Computational Science and Engineering*, Amsterdam, Netherlands, February 2023.

**Unified analysis of discontinuous Galerkin and flux reconstruction methods based on the summation-by-parts property.** *International Conference on Spectral and High-Order Methods* (online), July 2021.

**Unified analysis of high-order methods based on the summation-by-parts property: Application to discontinuous Galerkin and flux reconstruction discretizations.** *SIAM Conference on Computational Science and Engineering* (online), March 2021.

## INVITED TALKS (SELECTED)

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**Mass-conservative adaptive high-order methods for the Richards equation.** *PIMS Research Workshop on Structure-Preserving Discretizations and their Applications*, Regina, Canada, June 2026.

**Extending Trixi.jl to PDEs on manifolds.** *Trixi User and Developer Interaction (TRUDI)*, Cologne, Germany, March 2026.

**Split-form and entropy-stable high-order methods for hyperbolic systems on curved unstructured meshes and manifolds.** *University of Ottawa Applied Mathematics Seminar*, Ottawa, Canada, November 2025.

**Robust split-form and entropy-stable high-order methods for next-generation atmospheric dynamics.** Canadian Meteorological Centre, Environment and Climate Change Canada, Dorval, Canada, August 2025.

**An entropy-stable discontinuous spectral-element method for the spherical shallow water equations in covariant form.** *University of Waterloo Numerical Analysis and Scientific Computing Seminar*, Waterloo, Canada, February 2025.

**Provably stable tensor-product discontinuous spectral-element methods on triangular and tetrahedral unstructured grids.** Ansys, Inc. (online), July 2024.

**Entropy-stable tensor-product discontinuous spectral-element methods on curved triangles and tetrahedra.** Rice University (online), March 2024.

**Efficient and robust spectral-element methods on triangles using tensor-product summation-by-parts operators.** *NASA Advanced Modeling and Simulation Seminar Series* (online), October 2022.

**Robust deformation of unstructured grids.** Bombardier Aerospace, Montréal, Canada, August 2016.

## PROFESSIONAL EXPERIENCE

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**University of Saskatchewan** Saskatoon, Canada (remote)  
*Postdoctoral Fellow, Department of Computer Science* January 2026 – Present  
**Advisor:** Raymond Spiteri

**Nonlinear Numerics Inc.** Ottawa, Canada  
*President* May 2025 – Present  
Specialized research, software development, and consulting services for scientific and engineering simulation; completed an industry contract developing a GPU-accelerated discontinuous Galerkin particle-in-cell solver for plasma simulations

**University of Cologne** Cologne, Germany (remote)  
*Research Fellow, Department of Mathematics and Computer Science* January 2025 – January 2026  
Academic research and student mentorship on high-order methods for atmospheric dynamics

**University of Cologne** Cologne, Germany  
*Postdoctoral Researcher, Department of Mathematics and Computer Science* January 2024 – January 2025  
**Project:** Robust structure-preserving discontinuous spectral-element methods for the ICON-DG atmospheric dynamical core  
**Advisor:** Gregor Gassner

**National University of Singapore** Singapore  
*Visiting Researcher, Department of Mechanical Engineering* August 2022  
**Project:** Koopman operator approaches to data-driven stability analysis of numerical methods  
**Advisor:** Gianmarco Mengaldo

**University of Toronto** Toronto, Canada  
*Graduate Researcher, Institute for Aerospace Studies* September 2018 – December 2023  
**Project:** Provably stable discontinuous spectral-element methods with the summation-by-parts property for conservation laws on general unstructured grids  
**Advisor:** David Zingg

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| <b>University of Toronto</b><br><i>Undergraduate Researcher, Institute for Aerospace Studies</i><br><b>Projects:</b> Optimization of finite-difference operators with the summation-by-parts property on non-uniform nodal distributions; comparison of continuation methods for steady aerodynamic flows<br><b>Advisor:</b> David Zingg      | Toronto, Canada<br>May 2017 – August 2017  |
| <b>McGill University</b><br><i>Undergraduate Researcher, Department of Mechanical Engineering</i><br><b>Project:</b> Robust deformation of unstructured grids using radial basis functions and linear elasticity<br><b>Advisor:</b> Siva Nadarajah                                                                                            | Montréal, Canada<br>May 2016 – August 2016 |
| <b>Carleton University</b><br><i>Undergraduate Researcher, Department of Mechanical and Aerospace Engineering</i><br><b>Projects:</b> Optimal estimation of uncertain parameters in computer models of welding processes; novel interpolation techniques for stress and strain tensor fields using quaternions<br><b>Advisor:</b> John Goldak | Ottawa, Canada<br>May 2015 – August 2015   |

## TEACHING AND COURSE DEVELOPMENT

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| <b>HydroLearn</b><br><i>High-Performance Parallel Computing for Hydrologic Modeling</i><br>Co-lead for the development of an online course on best practices for high-performance parallel computing in hydrology as part of an open-source educational platform for hydrologic science and engineering                                                                                                                                                                                                                                                                                                 | <a href="https://hydrolearn.org/">https://hydrolearn.org/</a><br>May 2026 – Present |
| <b>CIROH Developers Conference</b><br><i>Orchestrating End-to-End Reproducible NextGen and NextGen-Adjacent Workflows</i><br>Co-organizer and presenter of an interactive in-person workshop on computational workflows for hydrologic simulations, covering the fundamentals of parallel computing and demonstrating parallelization strategies for hydrologic model execution and parameter estimation using the SYMFLUENCE software framework                                                                                                                                                        | Salt Lake City, USA<br>May 2026                                                     |
| <b>University of Cologne</b><br><i>Scientific Computing: Introduction to Atmospheric Flow Modeling (14722.0023)</i><br>Assisted in the development and delivery of a new graduate-level mathematics course on modern numerical methods for atmospheric simulations; contributed to curriculum design and preparation of lecture notes, and was responsible for the creation and supervision of the final project, in which students implement a discontinuous Galerkin solver for the spherical shallow water equations and assess its effectiveness for a series of standard atmospheric test problems | Cologne, Germany<br>April 2024 – July 2024                                          |

## SUPERVISION OF GRADUATE AND UNDERGRADUATE RESEARCH

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| <b>Fabian Höck, University of Cologne</b><br><i>MSc, Mathematics (co-supervised with Gregor Gassner and Benedict Geihe)</i><br><b>Project:</b> A discontinuous Galerkin method for moist atmospheric dynamics with rain                                      | Cologne, Germany<br>April 2024 – April 2025       |
| <b>Ruilin (Jerry) Bai, University of Toronto</b><br><i>BASc, Engineering Science (co-supervised with David Zingg)</i><br><b>Project:</b> Optimization of summation-by-parts operators for minimal solution error                                             | Toronto, Canada<br>September 2020 – December 2020 |
| <b>Yewon Lee, University of Toronto</b><br><i>BASc, Engineering Science (co-supervised with David Zingg and Masayuki Yano)</i><br><b>Project:</b> Comparative evaluation of energy- and entropy-stable discontinuous Galerkin and summation-by-parts methods | Toronto, Canada<br>May 2019 – August 2019         |

## AWARDS AND SCHOLARSHIPS

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| NSERC Canada Postdoctoral Research Award                                     | 2026 – 2028              |
| Best Student Paper, International Conference on Computational Fluid Dynamics | 2022                     |
| Ontario Graduate Scholarship                                                 | 2019 – 2020, 2022 – 2023 |
| Kenneth Molson Fellowship                                                    | 2021 – 2022, 2022 – 2023 |
| Queen Elizabeth II Graduate Scholarship in Science and Technology            | 2020 – 2021, 2021 – 2022 |
| Douglas Patton Hogg Memorial Award                                           | 2021                     |
| NSERC Canada Graduate Scholarship - Master's                                 | 2018 – 2019              |

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| University Medal (highest academic standing of any Carleton engineering graduate) | 2018        |
| Canadian Society for Mechanical Engineering Gold Medal                            | 2018        |
| Rajesh Ahluwalia Memorial Scholarship                                             | 2017 – 2018 |
| NSERC Undergraduate Student Research Award                                        | 2015, 2017  |
| McGill University Summer Undergraduate Research in Engineering Award              | 2016        |
| Allan Buchanan Undergraduate Scholarship                                          | 2015 – 2016 |
| Deans' Honour List, Carleton University                                           | 2014 – 2018 |
| Faculty Scholarship, Carleton University                                          | 2014 – 2018 |

## OPEN-SOURCE SOFTWARE CONTRIBUTIONS AS PRINCIPAL DEVELOPER

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| <b>TrixiAtmo.jl</b>                                                                                                                                                                                                                                                    | <a href="https://github.com/trixi-framework/TrixiAtmo.jl">https://github.com/trixi-framework/TrixiAtmo.jl</a>                         |
| Julia package extending the <a href="#">Trixi.jl numerical framework</a> for conservation laws to enable the solution of atmospheric flow problems using a high-order discontinuous spectral-element dynamical core                                                    |                                                                                                                                       |
| <b>HydroTrixi.jl</b>                                                                                                                                                                                                                                                   | <a href="https://github.com/uofs-simlab/HydroTrixi.jl">https://github.com/uofs-simlab/HydroTrixi.jl</a>                               |
| Downstream package extending <a href="#">Trixi.jl</a> to nonlinear parabolic systems with storage terms, with a focus on applications in hydrologic science and engineering                                                                                            |                                                                                                                                       |
| <b>StableSpectralElements.jl</b>                                                                                                                                                                                                                                       | <a href="https://github.com/tristanmontoya/StableSpectralElements.jl">https://github.com/tristanmontoya/StableSpectralElements.jl</a> |
| Julia framework for energy-stable and entropy-stable discontinuous spectral-element methods on general element types based on multidimensional and tensor-product formulations; emphasis on dispatched strategies for matrix-based and matrix-free operator evaluation |                                                                                                                                       |
| <b>GHOST: Generalized High-Order Solver Toolbox</b>                                                                                                                                                                                                                    | <a href="https://github.com/tristanmontoya/GHOST">https://github.com/tristanmontoya/GHOST</a>                                         |
| Python implementation of discontinuous Galerkin and flux reconstruction schemes in one or two spatial dimensions with various design choices                                                                                                                           |                                                                                                                                       |

## OTHER SIGNIFICANT OPEN-SOURCE CONTRIBUTIONS

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| <b>Trixi.jl</b>                                                                                                                                                              | <a href="https://github.com/trixi-framework/Trixi.jl">https://github.com/trixi-framework/Trixi.jl</a> |
| Member of the core development team since 2024; contributed support for PDEs on manifolds, purely parabolic equations, and save/load functionality for general element types |                                                                                                       |

## SERVICE

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| <b>Centre for Computational Science and Engineering</b>                                                                                                                                                                                                | Toronto, Canada                                               |
| <i>Lab Representative, Computational Aerodynamics</i>                                                                                                                                                                                                  | <i>December 2021 – June 2024</i>                              |
| Member of the student organizing committee for an interdepartmental group of researchers at the University of Toronto across various disciplines of computational science and engineering; assisted in hosting the <a href="#">2023 CCSE Symposium</a> |                                                               |
| <b>Canadian Science Fair Journal</b>                                                                                                                                                                                                                   | <a href="https://csfjournal.com/">https://csfjournal.com/</a> |
| <i>Editor, Physics and Mathematics Section</i>                                                                                                                                                                                                         | <i>June 2022 – March 2024</i>                                 |
| Volunteer editor and mentor for an open-access online publication showcasing science projects by children and youth at primary and secondary schools across Canada                                                                                     |                                                               |
| <b>Reviewer for Scientific Journals and Conferences</b>                                                                                                                                                                                                |                                                               |
| Journal of Parallel and Distributed Computing, Advances in Continuous and Discrete Models, JuliaCon                                                                                                                                                    |                                                               |

## TECHNICAL SKILLS

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| <b>Programming languages</b>      | <b>Daily:</b> Julia, C/C++, Python, $\LaTeX$ ; <b>Occasional:</b> Fortran, MATLAB, HTML/CSS                                                                                                                                    |
| <b>Development tools</b>          | Unix shell/Bash scripting, Git, CMake/CTest, GNU Make, Jupyter/Pluto notebooks, agentic workflows (OpenAI Codex, Claude Code), continuous integration/deployment (GitHub Actions), containerization (Docker), VS Code, Vim     |
| <b>High-performance computing</b> | Shared-memory parallelism (OpenMP, multithreading), distributed-memory parallelism (MPI), task/actor-based concurrency (CAF), GPU acceleration (CUDA), SIMD/SIMT vectorization, matrix-free algorithms, job scheduling (Slurm) |

## OTHER ACTIVITIES

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Music (performing guitarist/bassist), cycling, alpine skiing (former ski instructor), non-fiction reading